

Lab Title:- WEP40 WIRELESS PACKET DECRYPTION AND AIRCRAK FORENSICS

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Course Name:- BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

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Executive summary

The covers lab covers some different aspect of WEP40 wireless packet/aircrack analysis such as decryption/identification of WEP40 wireless packet/aircrack and also extraction/recovering of object in the packet.

Lab Objectives

The objectives of the lab is to understand how to conduct some basic WEP40 wireless packet/aircrack packet forensic.

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

- tshark:- Dump and analyze network traffic.
 - wireshark:- is a GUI network protocol analyzer that lets you interactively browse packet data from a live network or from a previously saved capture file.
 - sha256sum:- Print/compute or check SHA256 (256-bit) checksums message digest.
 - Aircrack-ng :- is an 802.11 WEP, 802.11i WPA/WPA2, and 802.11w WPA2 key cracking program.
 - File:- used to determine file type
 - unxz:- is a general-purpose data compression/decompression tool with command line syntax similar to gzip(1) and bzip2(1)
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Methology

The lab was conducted by following the practical step by step procedure provided in the lab assessment document/practical manual powerpoint slide

Screenshot and Evidence

SCREENSHOT SHOWING THE COMPRESSED EVIDENCE FILE AND HASH

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ wget -O evidence/file.xz \
'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
--2026-09-24 19:09:07-- https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: 'evidence/file.xz'

evidence/file.xz          100%[=====>] 12.24M  3.76MB/s  in 3.6s

2026-09-24 19:09:14 (3.43 MB/s) - 'evidence/file.xz' saved [12837660/12837660]

(kali@kali)-[~/SBT-DF203-Lab9]
$ sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  evidence/file.xz

(kali@kali)-[~/SBT-DF203-Lab9]
```

SCREENSHOT SHOWING THE DECOMPRESSED CAPTURE IDENTIFICATION

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ unxz -k workings/file_working.xz

(kali@kali)-[~/SBT-DF203-Lab9]
$ file workings/file_working
workings/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)

(kali@kali)-[~/SBT-DF203-Lab9]
```

SCREENSHOT SHOWING THE PROTOCOL HIERARCHY

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r workings/file_working -q -z io,phs | tee reports/protocol_hierarchy.txt

Protocol Hierarchy Statistics
Filter:

frame  wlan  data  frames:45169 bytes:13511274
      wlan  frames:45169 bytes:13511274
      data  frames:15713 bytes:13104128
```

SCREENSHOT SHOWING THE 802.11 FRAME CATEGORIES

```
(kali@kali)-[/SBT-DF203-Lab9]
$ tshark -r workings/file_working -Y 'wlan' -T fields \
  -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
  | head -n 100 | tee reports/wlan_frame_sample.tsv
```

frame.number	frame.time	wlan.fc.type	wlan.fc.subtype	wlan.sa	wlan.da	wlan.bssid
1	2015-03-06T02:51:54.364526000-0500	1	11			
2	2015-03-06T02:51:54.364527000-0500	1	11			
3	2015-03-06T02:51:54.364525000-0500	1	11			
4	2015-03-06T02:51:54.364526000-0500	1	11			
5	2015-03-06T02:51:54.365550000-0500	1	11			
6	2015-03-06T02:51:54.366063000-0500	1	11			
7	2015-03-06T02:51:54.366062000-0500	1	11			
8	2015-03-06T02:51:54.366061000-0500	1	11			
9	2015-03-06T02:51:54.366063000-0500	1	11			
10	2015-03-06T02:51:54.366061000-0500	1	11			
11	2015-03-06T02:51:54.366064000-0500	1	11			
12	2015-03-06T02:51:54.366063000-0500	1	11			
13	2015-03-06T02:51:54.366063000-0500	1	11			
14	2015-03-06T02:51:54.366063000-0500	1	11			
15	2015-03-06T02:51:54.366064000-0500	1	11			
16	2015-03-06T02:51:54.366062000-0500	1	11			
17	2015-03-06T02:51:54.366575000-0500	1	11			
18	2015-03-06T02:51:54.366575000-0500	1	11			
19	2015-03-06T02:51:54.366573000-0500	1	11			
20	2015-03-06T02:51:54.366574000-0500	1	11			
21	2015-03-06T02:51:54.366575000-0500	1	11			
22	2015-03-06T02:51:54.366575000-0500	1	11			
23	2015-03-06T02:51:54.366575000-0500	1	11			
24	2015-03-06T02:51:54.366574000-0500	1	11			
25	2015-03-06T02:51:54.366574000-0500	1	11			
26	2015-03-06T02:51:54.366574000-0500	1	11			
27	2015-03-06T02:51:54.366574000-0500	1	11			
28	2015-03-06T02:51:54.366574000-0500	1	11			
29	2015-03-06T02:51:54.366574000-0500	1	11			
30	2015-03-06T02:51:54.366575000-0500	1	11			
31	2015-03-06T02:51:54.366573000-0500	1	11			
32	2015-03-06T02:51:54.366575000-0500	1	11			

SCREENSHOT SHOWING THE PROTECTED WEP FRAMES AND IV EVIDENCE

```
(kali@kali)-[/SBT-DF203-Lab9]
$ tshark -r workings/file_working -Y 'wlan.fc.protected==1' -T fields \
  -e frame.number -e frame.time epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
  | head -n 200 | tee reports/wep_protected_frames.tsv
```

frame.number	frame.time epoch	wlan.sa	wlan.da	wlan.bssid	wlan.wep.iv	wlan.wep.key
9432	1425628319.232982000	00:26:66:55:97:d4	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0x67ffec	0
10130	1425628320.138252000	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	00:26:66:55:97:d6	0xc2cbeb	0
10155	1425628320.176656000	f0:f6:1c:68:96:7c	04:8d:38:48:e3:b4	00:26:66:55:97:d6	0xc3cbeb	0
10157	1425628320.177168000	f0:f6:1c:68:96:7c	04:8d:38:48:a8:b5	00:26:66:55:97:d6	0xc4cbeb	0
10159	1425628320.177168000	f0:f6:1c:68:96:7c	08:10:77:92:7c:2f	00:26:66:55:97:d6	0xc5cbeb	0
10161	1425628320.177168000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0xc6cbeb	0
10163	1425628320.177168000	f0:f6:1c:68:96:7c	c8:3a:35:58:c4:ba	00:26:66:55:97:d6	0xc7cbeb	0
10165	1425628320.177167000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc8cbeb	0
10167	1425628320.187417000	00:26:66:55:97:d4	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc9ffec	0
10170	1425628320.180760000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc9ffec	0
10171	1425628320.182288000	f0:f6:1c:68:96:7c	33:33:00:00:00:02	00:26:66:55:97:d6	0xc9cbeb	0
10172	1425628320.187417000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc9cbeb	0
10173	1425628320.187408000	f0:f6:1c:68:96:7c	33:33:00:00:00:02	00:26:66:55:97:d6	0xc9cbeb	0
10175	1425628320.187919000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xcacbeb	0
10176	1425628320.188952000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc9ffec	0
10178	1425628320.189976000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc9ffec	0
10183	1425628320.192016000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc9cbeb	0
10185	1425628320.193848000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc9ffec	0
10224	1425628320.259088000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc9cbeb	0
10228	1425628320.262680000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xc9ffec	0
10229	1425628320.262078000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc9ffec	0
10233	1425628320.263184000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0xc9cbeb	0
10261	1425628320.419350000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0xc9cbeb	0
10266	1425628320.420374000	f0:f6:1c:68:96:7c	01:00:5e:00:00:02	00:26:66:55:97:d6	0xc9cbeb	0
10268	1425628320.420374000	f0:f6:1c:68:96:7c	01:00:5e:00:00:fb	00:26:66:55:97:d6	0xd0cbeb	0
10270	1425628320.420886000	f0:f6:1c:68:96:7c	01:00:5e:00:00:02	00:26:66:55:97:d6	0x70ffec	0
10272	1425628320.422421000	f0:f6:1c:68:96:7c	01:00:5e:00:00:fb	00:26:66:55:97:d6	0x71ffec	0
10353	1425628320.583190000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0x72ffec	0
10355	1425628320.583700000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0x73ffec	0
10357	1425628320.583702000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xd1cbeb	0
10359	1425628320.584726000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0x74ffec	0
10360	1425628320.586774000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0xd2cbeb	0
10367	1425628320.789014000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0xd3cbeb	0
10375	1425628320.936121000	f0:f6:1c:68:96:7c	ff:ff:ff:ff:ff:ff	00:26:66:55:97:d6	0xd4cbeb	0
11400	1425628340.483347000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0xbaffec	0
11402	1425628340.483347000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x17cbeb	0
11406	1425628340.490003000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x18cbeb	0
11408	1425628340.491538000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x19cbeb	0
11410	1425628340.559638000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xb0ffec	0
11412	1425628340.562706000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x1bcbeb	0
11414	1425628340.655382000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xbcffec	0
11416	1425628340.655382000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xbdffec	0
11418	1425628340.656405000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xb0ffec	0
11420	1425628340.656405000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xb0ffec	0
11422	1425628340.656917000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc0ffec	0
11424	1425628340.657430000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc2ffec	0
11428	1425628340.657942000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc3ffec	0
11430	1425628340.657942000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc4ffec	0
11432	1425628340.658454000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc5ffec	0
11434	1425628340.658962000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x1ccbeb	0
11436	1425628340.658962000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x1dcbeb	0
11438	1425628340.658962000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x1ecbeb	0
11440	1425628340.662034000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x1fcbeb	0
11442	1425628340.662035000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x20cbeb	0
11444	1425628340.729862000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc6ffec	0
11446	1425628340.740882000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x21cbeb	0
11448	1425628340.747538000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x22cbeb	0
11450	1425628340.818197000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc7ffec	0
11452	1425628340.818198000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc8ffec	0
11454	1425628340.818198000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc9ffec	0
11456	1425628340.818198000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc9ffec	0
11458	1425628340.819222000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc9ffec	0
11460	1425628340.819222000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc9ffec	0
11462	1425628340.819734000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc9ffec	0
11464	1425628340.819734000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc9ffec	0
11466	1425628340.820243000	f0:f6:1c:68:96:7c	00:26:66:55:97:d4	00:26:66:55:97:d6	0x23cbeb	0
11468	1425628340.821270000	00:26:66:55:97:d4	f0:f6:1c:68:96:7c	00:26:66:55:97:d6	0xc9ffec	0

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r workings/file_working -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
  | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
  7 0x42e8eb
  7 0x28d3eb
  6 0x95dfef
  5 0x9415ed
  5 0x86e5eb
  5 0x7ae5eb
  5 0x28e7eb
  5 0x19e5eb
  4 0xd5d7eb
  4 0xd3e8eb
  4 0xc915ed
  4 0xc7d8eb
  4 0xade4eb
  4 0x73e5eb
  4 0x65d8eb
  4 0x4ad5eb
  4 0x31e7eb
  4 0x2723ed
  4 0x10e7eb
  3 0xffe5eb
  3 0xf609ed
  3 0xa307ed
  3 0x97dbeb
  3 0x96d1eb
  3 0x92e8eb
  3 0x91e8eb
  3 0x8adeeb
  3 0x630aed
  3 0x56e8eb
  3 0x3c1aed
```

SCREENSHOT SHOWING THE AIRCRACK-NG RESULT

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ aircrack-ng workings/file_working | tee reports/aircrack_output.txt
Reading packets, please wait...
Opening workings/file_working
Read 45169 packets.

# BSSID      ESSID      Encryption
1 00:26:66:55:97:D6  cgnetwork  WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait...
Opening workings/file_working
Read 45169 packets.

1 potential targets

Attack will be restarted every 5000 captured ivs.

Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB  depth  byte(vote)
0   0/ 1    A4(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456) 91(19200) B7(19200) 26(18944) 68(18944) 9E(18944) 0E(18688) 74(18688)
1   0/ 1    3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712) B0(19712) BD(19456) 80(19200) 85(19200) 9D(19200) 13(18944) EC(18944)
2   0/ 1    F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456) 0C(19200) 45(18944) 4F(18944) A8(18944) F7(18944) 60(18688) 63(18688)
3   1/ 10   F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456) B2(19456) 09(19456) 20(19456) 8F(19200) A0(19200) B0(19200) 04(18944)
4   6/ 9    01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456) F9(19456) 45(18944) 85(18944) 95(18944) D3(18944) 3A(18688) 40(18688)

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%
```

SCREENSHOT SHOWING THE AIRDECAP-NG DECRYPTION RESULT

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
A4:3D:F6:F3:74

(kali@kali)-[~/SBT-DF203-Lab9]
$ cd workings

(kali@kali)-[~/SBT-DF203-Lab9/workings]
$ airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets    0
Number of decrypted WPA packets    0
Number of bad TKIP (WPA) packets   0
Number of bad CCMP (WPA) packets   0

(kali@kali)-[~/SBT-DF203-Lab9/workings]
$
```

SCREENSHOT SHOWING THE DECRYPTED IP/MAC ENDPOINT INVENTORY

```
(kali@kali)~[/SBT-DF203-Lab9]
$ tshark -r "$(find workings -maxdepth 1 -type f -name '*-dec*' | head -n 1)" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt

Ethernet Endpoints
Filter:<No Filter>

| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
Apple_68:96:7c 15407 12 MB 7076 616 kB 8331 11 MB
EFMNetworks_55:97:d4 15020 12 MB 8178 11 MB 6842 593 kB
ASUSTekCOMPU_2a:c2:7a 358 281 kB 223 270 kB 135 11 kB
IPv4mcast_fb 42 6,648 bytes 0 0 bytes 42 6,648 bytes
IPv6mcast_fb 36 7,022 bytes 0 0 bytes 36 7,022 bytes
Broadcast 31 3,386 bytes 0 0 bytes 31 3,386 bytes
IPv4mcast_7f:ff:fa 31 11 kB 0 0 bytes 31 11 kB
IPv6mcast_16 10 920 bytes 0 0 bytes 10 920 bytes
IPv6mcast_02 9 630 bytes 0 0 bytes 9 630 bytes
IPv4mcast_16 3 180 bytes 0 0 bytes 3 180 bytes
IPv4mcast_02 2 92 bytes 0 0 bytes 2 92 bytes
NetisTechnol_48:e5:b4 1 42 bytes 0 0 bytes 1 42 bytes
NetisTechnol_48:a8:b5 1 42 bytes 0 0 bytes 1 42 bytes
08:10:77:92:7c:2f 1 42 bytes 0 0 bytes 1 42 bytes
TendaTechnol_58:c4:ba 1 42 bytes 0 0 bytes 1 42 bytes
IPv6mcast_ff:2a:c2:7a 1 78 bytes 0 0 bytes 1 78 bytes

(kali@kali)~[/SBT-DF203-Lab9]

(kali@kali)~[/SBT-DF203-Lab9]
$ tshark -r "$(find workings -maxdepth 1 -type f -name '*-dec*' | head -n 1)" -q -z endpoints,ip | tee reports/ip_endpoints.txt

IPv4 Endpoints
Filter:<No Filter>

| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
192.168.0.15 15309 12 MB 6995 608 kB 8314 11 MB
198.90.20.111 8057 6,916 kB 4446 6,637 kB 3611 279 kB
199.27.79.193 5694 4,825 kB 3131 4,603 kB 2563 221 kB
192.168.0.9 329 277 kB 197 265 kB 132 11 kB
173.194.127.141 153 109 kB 80 102 kB 73 7,364 bytes
173.194.127.55 143 89 kB 71 82 kB 72 6,990 bytes
8.8.8.8 81 10 kB 40 6,960 bytes 41 3,391 bytes
199.16.158.186 66 15 kB 29 10 kB 37 5,214 bytes
173.194.127.227 66 24 kB 26 15 kB 40 8,455 bytes
173.194.127.229 55 22 kB 23 13 kB 32 9,731 bytes
31.13.70.1 51 19 kB 26 15 kB 25 3,151 bytes
173.241.248.220 44 6,848 bytes 20 3,122 bytes 24 3,726 bytes
198.41.208.140 44 23 kB 21 25 kB 23 2,718 bytes
224.0.0.251 42 6,648 bytes 0 0 bytes 42 6,648 bytes
173.194.127.237 36 9,526 bytes 14 6,435 bytes 22 3,091 bytes
216.58.220.234 34 15 kB 16 12 kB 18 2,497 bytes
192.168.0.1 33 12 kB 33 12 kB 0 0 bytes
173.241.248.219 33 5,733 bytes 14 2,807 bytes 19 2,926 bytes
104.74.158.12 33 21 kB 18 20 kB 15 1,230 bytes
74.125.203.156 32 20 kB 16 17 kB 16 2,534 bytes
239.255.255.250 31 11 kB 0 0 bytes 31 11 kB
173.194.127.231 31 6,816 bytes 12 5,197 bytes 19 1,619 bytes
198.41.209.137 28 16 kB 13 14 kB 15 1,556 bytes
17.172.239.31 25 6,504 bytes 12 3,973 bytes 13 2,531 bytes
173.194.127.95 23 4,077 bytes 9 2,232 bytes 14 1,845 bytes
173.194.127.122 21 12 kB 11 11 kB 10 1,759 bytes
69.59.169.185 18 5,644 bytes 7 3,675 bytes 11 1,969 bytes
72.246.103.16 16 3,697 bytes 6 1,732 bytes 10 1,965 bytes
23.23.127.192 15 3,383 bytes 6 1,370 bytes 9 2,013 bytes

(kali@kali)~[/SBT-DF203-Lab9]
$ tshark -r "$(find workings -maxdepth 1 -type f -name '*-dec*' | head -n 1)" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.co
l.Protocol \
| head -n 200 | tee reports/ip_mac_mapping_sample.tsv
7 0.0.0.0 255.255.255.255 DHCP
9 0.0.0.0 255.255.255.255 DHCP
11 192.168.0.1 255.255.255.255 DHCP
20 192.168.0.15 8.8.8.8 DNS
21 192.168.0.15 8.8.8.8 DNS
22 192.168.0.15 224.0.0.2 IGMPv2
23 192.168.0.15 224.0.0.251 IGMPv2
24 192.168.0.15 224.0.0.2 IGMPv2
25 192.168.0.15 224.0.0.251 IGMPv2
26 8.8.8.8 192.168.0.15 DNS
27 8.8.8.8 192.168.0.15 DNS
30 192.168.0.15 17.172.239.31 TCP
31 192.168.0.15 17.172.239.127 TCP
33 17.172.239.31 192.168.0.15 TCP
35 192.168.0.15 17.172.239.31 TCP
36 192.168.0.15 17.172.239.31 TLSv1
37 17.172.239.127 192.168.0.15 TCP
38 192.168.0.15 17.172.239.127 TCP
42 17.172.239.31 192.168.0.15 TCP
43 17.172.239.31 192.168.0.15 TLSv1
44 17.172.239.31 192.168.0.15 TLSv1
45 17.172.239.31 192.168.0.15 TLSv1
47 0.0.0.0 255.255.255.255 DHCP
48 0.0.0.0 255.255.255.255 DHCP
49 192.168.0.15 17.172.239.31 TCP
50 192.168.0.15 17.172.239.31 TCP
51 192.168.0.1 192.168.0.15 DHCP
52 192.168.0.15 17.172.239.31 TLSv1
55 17.172.239.31 192.168.0.15 TCP
```

SCREENSHOT SHOWING THE EXPORTED IMAGE OR HTML OBJECTS

```
(kali@kali)~/SBT-DF203-Lab9
$ tshark -r "$(find workings -maxdepth 1 -type f -name '*-dec*' | head -n 1)" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
1 0.000000 EFMNetworks 55:97:d4 → Broadcast ARP 42 Who has 192.168.0.15? Tell 192.168.0.1
2 0.943674 Apple 68:96:7c → NetisTechnol 48:e5:b4 ARP 42 Who has 192.168.0.1? Tell 192.168.0.141
3 0.944186 Apple 68:96:7c → NetisTechnol 48:a8:b5 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
4 0.944186 Apple 68:96:7c → 08:10:77:92:7c:2f ARP 42 Who has 192.168.0.1? Tell 192.168.0.12
5 0.944186 Apple 68:96:7c → EFMNetworks 55:97:d4 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
6 0.944186 Apple 68:96:7c → TendaTechnol 58:c4:ba ARP 42 Who has 192.168.0.1? Tell 192.168.0.113
7 0.944185 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0*216be1c
8 0.944784 EFMNetworks 55:97:d4 → Apple 68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
9 0.944778 192.168.0.15 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0*216be1c
10 0.949306 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
11 0.954435 192.168.0.1 → 255.255.255.255 DHCP 590 DHCP NAK - Transaction ID 0*216be1c
12 0.954937 Apple 68:96:7c → Broadcast ARP 42 ARP Announcement for 192.168.0.15
13 0.955970 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
14 0.956994 Apple 68:96:7c → Broadcast ARP 42 ARP Announcement for 192.168.0.15
15 0.959034 Apple 68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
16 0.960066 Apple 68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15

7 0.944185 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0*216be1c
8 0.944784 EFMNetworks 55:97:d4 → Apple 68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
9 0.944778 192.168.0.15 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0*216be1c
10 0.949306 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
11 0.954435 192.168.0.1 → 255.255.255.255 DHCP 590 DHCP NAK - Transaction ID 0*216be1c
12 0.954937 Apple 68:96:7c → Broadcast ARP 42 ARP Announcement for 192.168.0.15
13 0.955970 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
14 0.956994 Apple 68:96:7c → Broadcast ARP 42 ARP Announcement for 192.168.0.15
15 0.959034 Apple 68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
16 0.960066 Apple 68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15

17 1.358744 192.168.0.15 → 8.8.8.8 DNS 73 Standard query 809883 A www.apple.com
18 1.029698 Apple 68:96:7c → Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
19 1.029698 EFMNetworks 55:97:d4 → Apple 68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
20 1.030282 192.168.0.15 → 8.8.8.8 DNS 73 Standard query 809883 A www.apple.com
21 1.186368 192.168.0.15 → 8.8.8.8 DNS 85 Standard query 0xb640 A 48-courier.push.apple.com
22 1.187392 192.168.0.15 → 224.0.0.251 ICMPv2 46 Leave Group 224.0.0.251
23 1.187392 192.168.0.15 → 224.0.0.251 ICMPv2 46 Membership Report group 224.0.0.251
24 1.187392 192.168.0.15 → 224.0.0.251 ICMPv2 46 Leave Group 224.0.0.251
25 1.189439 192.168.0.15 → 224.0.0.251 ICMPv2 46 Membership Report group 224.0.0.251
26 1.358744 192.168.0.15 → 8.8.8.8 DNS 203 Standard query response 0*9503 A www.apple.com CNAME www.apple.com.edgekey.net CNAME e3191
.dscg.akamaiedge.net A 104.74.154.228
27 1.358744 192.168.0.15 → 8.8.8.8 DNS 299 Standard query response 0xb640 A 48-courier.push.apple.com CNAME 48-courier.push.apple.com.akadns.net CNAME us-courier.push-a
28 1.358744 192.168.0.15 → 8.8.8.8 DNS 299 Standard query response 0xb640 A 48-courier.push.apple.com CNAME 48-courier.push.apple.com.akadns.net CNAME us-courier.push-a
29 1.351744 Apple 68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
30 1.351744 TCP 68 → 192.168.0.15 [SYN] Seq=0 Win=0 Len=0 WS=16 TSV=549602888 TS=0 SACK_PERM
31 1.556032 192.168.0.15 → 17.172.239.127 TCP 78 50955 → 443 [SYN] Seq=0 Win=0 Len=0 MSS=1460 WS=16 TSV=549603081 TS=0 SACK_PERM
32 1.703489 Apple 68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
33 1.703489 17.172.239.31 → 192.168.0.15 TCP 74 443 → 50955 [SYN, ACK] Seq=0 Ack=1 Win=1460 Len=0 MSS=1460 SACK_PERM TSV=646109317 TS=549602888 WS=128
34 1.704513 Apple 68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
35 1.729889 192.168.0.15 → 17.172.239.31 TCP 66 50954 → 443 [ACK] Seq=1 Ack=1 Win=13768 Len=0 TSV=549603254 TS=549603254 TS=0 SACK_PERM
36 1.734325 192.168.0.15 → 17.172.239.31 TLSv1 255 Client Hello [SHA1-courier.push.apple.com]
37 1.879168 17.172.239.127 → 192.168.0.15 TCP 74 443 → 50955 [SYN, ACK] Seq=0 Ack=1 Win=1460 Len=0 MSS=1460 SACK_PERM TSV=641873548 TS=549603081 WS=128
38 1.906816 192.168.0.15 → 17.172.239.127 TCP 54 50955 → 443 [RST] Seq=1 Win=0 Len=0
39 1.921199 Apple 68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
40 1.924226 Apple 68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
41 2.054781 192.168.0.15 → 192.168.0.15 TCP 66 443 → 50954 [ACK] Seq=1 Ack=190 Win=15616 Len=0 TSV=646109707 TS=549603268
42 2.054784 17.172.239.31 → 192.168.0.15 TCP 66 443 → 50954 [ACK] Seq=1 Ack=190 Win=15616 Len=0 TSV=646109707 TS=549603268
43 2.054784 17.172.239.31 → 192.168.0.15 TLSv1 194 Server Hello [SHA1-courier.push.apple.com]
44 2.054784 17.172.239.31 → 192.168.0.15 TLSv1 1514 Certificate
45 2.054784 17.172.239.31 → 192.168.0.15 TLSv1 194 Server Key Exchange, Certificate Request, Server Hello Done
46 2.056834 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 90 Multicast Listener Report Message V2
47 2.088572 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0*216be1c
48 2.092161 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0*216be1c

(kali@kali)~/SBT-DF203-Lab9
$ foremost -i "$(find workings -maxdepth 1 -type f -name '*-dec*' | head -n 1)" -o exported/foremost | tee reports/foremost_log.txt
Processing: workings/file_working-dec
[*]

(kali@kali)~/SBT-DF203-Lab9
$ find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
exported/foremost/htm/00024427.htm: HTML document, ASCII text
exported/foremost/htm/00001639.htm: HTML document, ASCII text
exported/foremost/jpg/00000104.jpg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length 16, baseline, precision 8, 2492x2492, components 3
exported/foremost/jpg/00004538.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI), density 72x72, segment length 16, comment: "Lavc55.69.100", baseline, precision 8, 71
8x404, components 3
exported/foremost/gif/00010876.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00003719.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00010456.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00000406.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00003653.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00001260.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00010876.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00003719.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00010456.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00000406.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00003653.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00001260.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00002450.gif: GIF image data, version 89a, 1 x 1
exported/foremost/audit.txt: ASCII text
exported/foremost/png/00001765.png: PNG image data, 44 x 44, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00010671.png: PNG image data, 1 x 400, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00010214.png: PNG image data, 1 x 1, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00024242.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00001831.png: PNG image data, 12 x 12, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00020736.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/http_objects/blah.css: ASCII text
exported/http_objects/Figure-2.png: PNG image data, 13 x 274, 8-bit/color RGBA, non-interlaced
exported/http_objects/collapsible-widgets.css: ASCII text
exported/http_objects/collect3fvr16_v1336a=4420489678t-pageview=515d1=http3A%2F%2Fimgur.com%2F8iVyt66dr=http3A%2F%2Fm.reddit.com%2F6ul=en-us&de=UTF-8&dt=Epic%20Dogpile%2
0%20imgur&sd=32-bitesr=320x568vsr=980x1409je=06.utma=2473412.14055857036td=UA-6671908-2&=4400497: GIF image data, version 89a, 1 x 1
exported/http_objects/Figure-16.png: PNG image data, 230 x 144, 8-bit/color RGBA, non-interlaced
exported/http_objects/og1i5PA.gifv: HTML document, ASCII text
exported/http_objects/logo.png: PNG image data, 194 x 55, 8-bit/color RGBA, non-interlaced
exported/http_objects/css3fffamily=Oswald%7C: ASCII text
exported/http_objects/superfish.args.min.js: JavaScript source, ASCII text, with no line terminators
exported/http_objects/hit(2).json: JSON text data
exported/http_objects/Query-1.10.2.min.js: JavaScript source, ASCII text, with very long lines (32072)
exported/http_objects/object64961: data
exported/http_objects/hls3ifversion4.12: XML 1.0 document, ASCII text
exported/http_objects/header-rss.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-print.png: PNG image data, 65 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/homewave-upwhite.png: PNG image data, 1640 x 141, 8-bit/color RGB, non-interlaced
exported/http_objects/gavid=537200270res=320x568x326plg=qt%2Cpm6ch=UTF-8&tz=-5406ws=300x250: ASCII text
exported/http_objects/og1i5PA.mp4: data
exported/http_objects/share-linkedln.png: PNG image data, 87 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/share-twitter.png: PNG image data, 79 x 31, 8-bit/color RGBA, non-interlaced
exported/http_objects/object6589.application%2Fjavascript: ASCII text, with CR/LF line terminators
exported/http_objects/AmagVviewerOnline.js%3F1425580381: ASCII text, with very long lines (4586)
exported/http_objects/portfolio-slideshow.min.js: JavaScript source, ASCII text, with very long lines (897)
exported/http_objects/6i=196128llnocoookies=undefined: HTML document, ASCII text
exported/http_objects/rombo.png: PNG image data, 61 x 70, 8-bit/color RGBA, non-interlaced
exported/http_objects/activeviewer%3Fid=osdi&vavi=BLA0VQV35VNMMMTY9QWknIgcCADA_50u=GEAABABOAHIAQLgAgDIA5kE4AQBoAYCwhMDEIABt6i=16adk=109919536p=784,685,1034,9856tos=1163,0,0,0,0
6mtos=1163,1163,116,com%2Fads%3Fpage%3Dinside%26main%3D1%26hot%3D%26t1me%3D14256086366is=300,250&uc=5: GIF image data, version 89a, 1 x 1
exported/http_objects/object6491: data
exported/http_objects/css3fffamily=NotoSans%3A400%2C400italic%2C700%7CRoboto+Condensed%3A700italic%2C400%2C700%7CYanone+Kaffeesatz%3A400%2C700ver=3.8.1: ASCII text
```

SCREENSHOT SHOWING THE HASHES OF DECRYPTED CAPTURE AND RECOVERED FILES

```
(kali@kali)-[~/SBT-DF203-Lab9]
└─$ sha256sum workings/* | tee reports/all_working_file_hashes.txt
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  workings/file_working
167c91994c269777f9048227deb89882caf3c3c763977f2059604f9a6a40b04  workings/file_working-dec
dd54144caef34f228bf4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  workings/file_working.xz

(kali@kali)-[~/SBT-DF203-Lab9]
```

```
(kali@kali)-[~/SBT-DF203-Lab9]
└─$ find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
02e357aea1b284a9e6a710885744310191740a96d99bb189146b1cbc547ea79e  exported/foremost/htm/00024427.htm
1d11ae950ff1be55ca0e69a487315631958af8d295252ca8895c4e72b161eb  exported/foremost/htm/00001539.htm
3f9be0d32e04c1735b498dab0f138673e20591a5b39201c731a91dd180929ba6  exported/foremost/jpg/00000104.jpg
f891fa7ef7652a26a54638dd7b0aebc06067fd72ec42a2a636523a6371297dc  exported/foremost/jpg/00004539.jpg
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00010876.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00003719.gif
53f7e56783c414746a9fc3612637509560424617e3135248640eac142d6e0f94  exported/foremost/gif/00010456.gif
a0d3a0a0f7dc3bf32d176fc3dcdad6e7aba2867c4f4d1f7af6335d2cfc6c44f8  exported/foremost/gif/00000406.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00003653.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00001260.gif
ef1955ae757c8b966c83248350331bd3a30f658ced11f387f8ebf05ab3368629  exported/foremost/gif/00002450.gif
2630549e6bcab306aa8a55a2b6311154c194076ed364c461b0104c4c644aab86  exported/foremost/audit.txt
ee1c79d0b533ec5ff0acf199d84fbcd7dc7e07acbe9defcac65416b63687950d  exported/foremost/png/00001765.png
ddf5dd4e0ef2b185eb0b0af7b6e90ebe74a84384cb4700658e76e754c8bfe550  exported/foremost/png/00010671.png
ebf4f635a17d10d6eb46ba680b70142419aa3220f228001a036d311a22ee9d2a  exported/foremost/png/00010214.png
681fb2c21414bf0c7cedbbe1d57f9506424033bc89beed026e85ea6c85d7ccf  exported/foremost/png/00024242.png
534c09b9b2d436a30c260eaa82679d2a2c970d4962399a8da227ee07162a755  exported/foremost/png/00001831.png
792477b4d20b4cbf0580011cad06d792425ce4e2ec5d232c1b4b41e06ac09676  exported/foremost/png/00020736.png
3546b5559b78314d3d18d290751916a34bf5ca20a1b565cdef64274a7a1d59  exported/http_objects/blah.css
4fbb9f1faef95f7c4ceba5708d359307683fba641b78653b897c8484061f88  exported/http_objects/figure-2.png
3320608508ac4dc1752305089907642e8247d07fe95799482949f4a06e26287  exported/http_objects/collapsible-widgets.css
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/http_objects/collect33fv-16_v-j336a=4420489676t-pageview6_s=16dl=http%3A%2F%2Fimgur.com%2F8iVy7t66dr=
http%3A%2F%2Fm.reddit.com%2F8ul=en-us&de=UTF-8&dt=Epic%20Doggp1e%20-K20Imgur6sd=32-bit6sr=320x568&vp=980x14096je=06_utma=2473412.14055857036t1d=UA-6671908-26z=44004997
7697378c08c8b6e8279e4c8a83b95f1faf4e264a07e593c55b3e6753f5d1a0c6  exported/http_objects/figure-16.png
```

Conclusion

In conclusion this lab has laid a very strong foundation in understanding how to analyse wireless packet/ aircrack packet.

Reference

- ICDFA course Material (SBT-DF203(basic networking skills for digital forensics)- by COMMANDANT AMINU IDRIS
 - ICDFA Recorded sessions
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